01 · PROBLEM

Why won't ER ping-pong go away?

It's not a lack of beds — it's a lack of reliable data

2×
ERs declaring they can't accept patients
2023 → 2024

5.4%
Transfers taking over 30 min
up from 1.9% in 2023

Why it keeps happening

- Paramedics check hospitals **one phone call at a time** — even dispatch centers take **33 min** on average and contact **10 hospitals**
- The top reason for refusal isn't beds but **staff shortages (48.1%)** — a defensive call made without information

With reliable information, the first thing to drop is 'refuse because we don't know.'

People tried to fix it — but there was no reliable data

The government is speeding up allocation, but when we cross-checked the hospital data behind those decisions:

91.2% of hospitals that said 'cannot accept' actually had a specialist in that department

8.6 yrs Max age of stale bed values shown as real-time (25 hospitals)

0 → 4 Burn-center capacity — 0 found from a single source, 4 when cross-checked

Bed data is entered manually by staff. Having a specialist does not guarantee acceptance. Sources: National Medical Center, National Fire Agency, National Assembly; team measurements (E-Gen, HIRA).

02 · SO HERE'S HOW WE SOLVED IT

One call, evidence-backed candidates

Calls are organized automatically; hospital data is verified.

← Paramedic call · 1·2
Voice processing
Call → text
17 fields → 6 for matching

← Public data · HIRA · 3
Bed reliability
Cross-checks reported values
Probability valid now

Matching server · 4
Nearest hospital candidates · rule-based ranking
Approval status · temporary bed hold
Decision log · hash chain

↕ Matches · acceptance replies

Ambulance screen · Hospital screen
Candidates with reliability · acceptance replies

The matching server is the only server connected to the screens
For hospitals, pre-arrival calls stay on record as grounds for acceptance decisions

1 Voice → patient info · how it works



1 Transcribe Speech recognized in chunks at every pause

2 Read The whole call through a Korean encoder

3 Judge 15 item-specific judges, each focused on its own

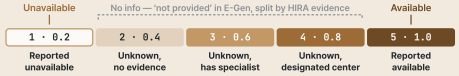
4 Organize Severity, chief complaint and vitals in one form

2 Bed reliability · how it works

1 Cross-check Reported values vs. HIRA specialist and specialty-hospital data



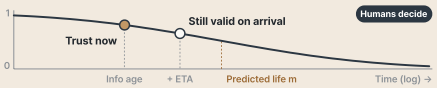
2 Grade 5 levels that separate 'reported unavailable' from 'no information'



3 Lifespan Learned from how each hospital has edited its values



4 Probability Probability it's correct now and on arrival — people make the call

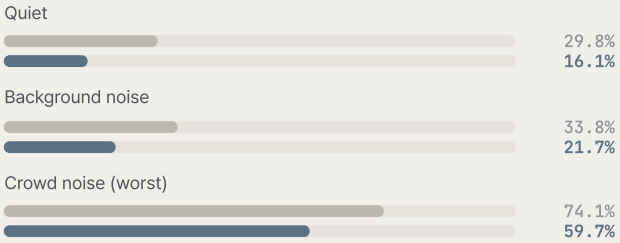


03 · PERFORMANCE — MEASURED OURSELVES

Results in numbers

1 Speech recognition · character error rate

500 sentences from real 119 calls — Before / After training



2 Patient info extraction

	Severity	Complaint	Per case
GoldenLink 0.35B	75%	97%	0.013 s
Generative 1.7B	78%	94%	6.9 s
External LLM API	61%	85%	0.4 s+

0.01 s with no data sent outside — accuracy on par with large models

3 Bed reliability validation

Live link to ERs nationwide ④	533
Merged with HIRA · median location error 11 m	97.2%
Time spent on stale data · same budget	−11%
Stale evidence in AI answers ⑤	33.3→23.3%

① 3 of 7 noise conditions shown ② 100 synthetic calls, same scoring ③ median of retrospective ER-bed simulation. ④ 533 = all ERs with live E-Gen bed feeds (as of OO) ⑤ AI answers n=OO, preliminary single-corpus result. All figures before human review.